higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER HARDWARE AND SOFTWARE
NQF LEVEL 3

(10041013)

12 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 A computer system's monitor is also known as the VDU.

1.2 The acronym PCB stands for personal computer board.

1.3 A computer's CMOS battery stores computer settings, such as date and time.

1.4 MemPerf is an open-source benchmark used to test memory bandwidth.

1.5 Software compatibility is a characteristic of software components or systems which can operate satisfactorily together on the same computer, or on different computers linked by a computer network.

1.6 Soft faults are difficult to diagnose compared to hard faults.

1.7 Milestones are short-term targets used to review progress.

1.8 Review procedures ensure that user requirements are not met.

1.9 Information log maintenance identifies sources for new information and trends for future reference.

1.10 Manufacturers should never be contacted when troubleshooting.

(10 × 1) **[10]**

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QUESTION 2

Choose a/an item/word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–O) next to the question number (2.1–2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Their purpose is to help identify faulty components	A	checklist
2.2	Helps to ensure that all aspects of the installation are covered	B	plug n play devices
2.3	Refers to the computer hardware that allows the operating system to detect it when first installed into a computer	C	diagnostic software applications
2.4	Ability of two or more computer components to work together	D	structure-based testing
2.5	Useful for understanding systems testing because it links the SDLC to the corresponding testing levels	E	northbridge chipset
2.6	Code is tested by means of checking the structure and dataflow	F	v-model
2.7	Manages high speed peripheral communications.	G	compatibility
2.8	Responsible for providing support to the onboard slower peripherals	H	upgrade installation
2.9	Allows for your operating system to be upgraded to a newer version	I	smart terminal
2.10	Guides you by providing a step-by-step installation process	J	Internet terminal
		K	motherboard
		L	software installation wizard
		M	southbridge chipset
		N	IP address
		O	software licence terms

(10 × 1) **[10]**

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QUESTION 3

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (3.1–3.10) in the ANSWER BOOK.

3.1 … fire is defined as a fire that uses electrical components.

A Class C
B Class B
C Class A
D Class D

3.2 This is a common source of a Class A fire.

A Wood and paper
B Titanium
C Grease
D Aluminium

3.3 Natural elements that are not good for electronic components include

A CMOS (complementary metal oxide semiconductor)
B Water, dust, heat and humidity
C Volcano, earthquakes and lightning
D ROM (read-only memory)

3.4 … is a tool used to troubleshoot the windows start-up process.

A loop
B interface
C msconfig
D tester

3.5 … is a symptom of a power-related problem.

A 1 short beep
B LED lights on the front panel not lighting up
C Blank screen and no beeping sound
D Coding

3.6 A contract between a service provider and an user or a company.

A SLA
B User sign-off
C Review procedure
D Third party

3.7 The … power-saving option saves all data in the RAM to the disk.

A Suspend
B Standby
C Hibernate
D Shut-down

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3.8 … combine the aspects of laptops and smart phones.

A Chromebooks
B Tablets
C Netbooks
D Personal digital assistants

3.9 A newer, faster laptop, which uses Google Chrome OS.

A Ultrabook
B Mecer
C Acer
D Chromebook

3.10 … produce high-quality text and images.

A Thermal printers
B Dot matrix printers
C Laser printers
D Switches

(10 × 1) **[10]**

TOTAL SECTION A: 30

SECTION B

QUESTION 4

4.1 Discuss the FIVE functional levels of a computer system. (5 × 2)

4.2 RAM or random-access memory is used in the PC for its primary memory. RAM is where all active programs and data are stored so that they are readily available and easily accessed by the CPU and other components of the PC. When you execute a program on your PC, a copy of the program is copied into RAM from whatever secondary storage it is on, usually the hard disk. Once it is in RAM, the instructions that make up the program are passed one at a time to the CPU for execution.

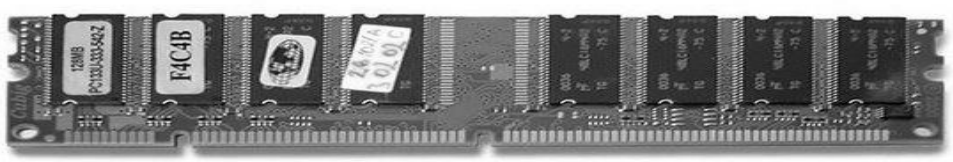


FIGURE 1

List FIVE different types of RAM and briefly explain the purpose of each type.
(5 × 2)
(2 × 10) **[20]**

QUESTION 5

5.1 Discuss TWO basic functions of a kernel. (2 × 2) (4)

5.2 List FIVE considerations that can help you estimate how much time will be required to complete PC hardware installation. (5)

5.3 Describe the steps you will take to make sure that system and application software communicate with the hardware. (5)

5.4 Write the acronym MFD out in full. (1)

[15]

QUESTION 6

6.1 List the FIVE tools needed to perform hardware installations. (5)

6.2 To keep your PC in good working condition, you should have a few simple tools and some cleaning supplies. The tools and supplies can be gathered from computer supply stores and even the grocery store.

Make a list of the tools and supplies you should have on hand to care for and maintain your PC. (7)

6.3 Before you open the system case and begin installing new memory modules in your PC, regardless of whether you are replacing existing modules or inserting additional memory, there are a few precautions you should take into consideration.

Briefly describe FIVE precautions for replacing and handling memory modules. (5)

[17]

QUESTION 7

7.1 Name SIX system features to be tested in order to check if the system works according to the specification.

7.2 Identify SIX planning risks and contingencies that may influence testing. (2 × 6) **[12]**

QUESTION 8

8.1 What does the acronym EULA stand for? (2)

8.2 Discuss the importance of EULA. (4 × 2) (8)

[10]

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QUESTION 9

9.1 John is trying to print out invitations for a birthday party, but the printer won't print. John approaches you for help and you have some ideas about what could be causing this problem.

Apply the process of elimination to solve John's problem.

9.2 If you have tried everything and the problem still is not solved, you have an option to escalate the problem to expert technical staff.

Describe the different types of reports that you will supply when forwarding undiagnosed faults to expert technical staff.

(2 × 5) **[10]**

QUESTION 10

10.1 Summarise the FIVE symptoms of technical computer problems. (5)

10.2 You have been employed as a helpdesk operator at IT Help Call Centre and your job is to assist a high number of customers as they call for help on daily basis. You are aware that the way you respond to user queries and problems is an important part of the image of the organisation.



Discuss the process of running a helpdesk in a call centre. (10)
[15]

QUESTION 11

A BIOS error message displayed on the monitor, regardless of the BIOS manufacturer, means that there is a serious problem with a system's FRM (field replaceable module) that is preventing the start-up of the computer.

11.1 Make a list of FIVE of the most common error messages displayed during the boot cycle and explain their meanings. (5 × 2) (10)

11.2 Formulate FIVE questions that you may ask the user to help pinpoint the cause of the problem. (5)

11.3 Mention SIX major features of multifunction devices. (6)
[21]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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